A SECTION OF A SMOOTH MORPHISM IS A REGULAR IMMERSION AND IS A RIGHT INVERSE OF THE FLAT PULLBACK BIVARIANT CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FF1 (sources/stacks-chow/pinned-original-source.tex line 13435).

Allowed standard prerequisites (editorial).

- A section of a smooth morphism is a regular immersion. Divisors, Lemma labelled divisors-lemma-section-smooth-regular-immersion (Stacks tag 067R), invoked at proof lines 13445-13446 and again at 13454-13456. Supplies both the first assertion of the statement and the regular embeddedness of the intersection used in the cycle computation.
- Flat pullback as a bivariant class. Same chapter, Lemma labelled lemma-flat-pullback-bivariant (source line 6131), invoked in the statement. Supplies the class $p^\frown *$ whose composite with $s^\frown !$ is asserted to be the identity.
- The fundamental-class identity for a regularly embedded closed subscheme. Same chapter, Lemma labelled lemma-gysin-fundamental (source line 12076), invoked at proof line 13457. Supplies the conclusion $s^\frown ! \text{ cap } p^\frown *[Z] = [Z]$ once the intersection is identified as a codimension r regular embedding.
- The base situation (S, delta) for operational Chow groups. Same chapter, Situation labelled situation-setup (source line 1372), invoked in the statement. Fixes the dimension function and the augmented ring $A^\frown *(X)$ in which the identity is stated.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma

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at source line 6131; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 067R: sources/stacks-chow/tags.txt, source lines 7980–7980. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13446 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBM: sources/stacks-chow/pinned-original-source.tex, source lines 12076–12076. Same-chapter pointer into the author environment lemma at source line 12076; invoked at proof line 13457 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $p : P \to X$ be a smooth morphism of schemes locally of finite type over S and let $s : X \to P$ be a section. Then s is a regular immersion and $1 = s^! \circ p^*$ in $A^*(X)^\wedge$ where $p^* \in A^*(P \to X)^\wedge$ is the bivariant class of Lemma 0B78.*

Proof. The first statement is Divisors, Lemma 067R. It suffices to show that $s^! \cap p^*[Z] = [Z]$ in $\mathrm{CH}_*(X)$ for any integral closed subscheme $Z \subset X$ as the assumptions are preserved by base change by $X' \to X$ locally of finite type. After replacing P by an open neighbourhood of $s(Z)$ we may assume $P \to X$ is smooth of fixed relative dimension r. Say $\dim_\delta(Z) = n$. Then every irreducible component of $p^{-1}(Z)$ has dimension $r + n$ and $p^*[Z]$ is given by $[p^{-1}(Z)]_{n+r}$. Observe that $s(X) \cap p^{-1}(Z) = s(Z)$ scheme theoretically. Hence by the same reference as used above $s(X) \cap p^{-1}(Z)$ is a closed subscheme regularly embedded in $\overline{p}^{-1}(Z)$ of codimension r. We conclude by Lemma 0FBM. □